Azure Virtual Networks Made Simple Beginner Guide

Azure Virtual Networks are the foundation of everything you build in the cloud, acting like a private digital fence for your servers. Learning how to create and manage them with Azure CLI will save you hours of manual clicking in the portal.

What you'll learn:

1. Creating Your First Virtual Network
2. Dividing Networks into Subnets
3. Securing Traffic with Security Groups
4. Connecting Two VNets with Peering

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Creating Your First Virtual Network

A Virtual Network (VNet) is your private space in Azure, similar to your home Wi-Fi network. It gives your virtual machines a secure place to talk to each other without touching the public internet. You define an IP address range (like 10.0.0.0/16) to decide how big your network is. Using Azure CLI makes creating this instant and repeatable.

```
$az network vnet create --resource-group MyResourceGroup --name MyVNet --address-prefix 10.0.0.0/16
```

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Dividing Networks into Subnets

A subnet is a smaller piece of your VNet, like dividing a large office building into separate rooms for different departments. You keep web servers in one subnet and database servers in another to keep things organized and secure. This allows you to apply different security rules to each section. It is a best practice to plan your subnet IP ranges carefully before creating them.

```
$az network vnet subnet create --resource-group MyResourceGroup --vnet-name MyVNet --name FrontendSub
```

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Securing Traffic with Security Groups

A Network Security Group (NSG) acts like a digital security guard standing at the door of your subnet or virtual machine. It filters incoming and outgoing network traffic using simple allow or deny rules based on IP address and port numbers. For example, you can open port 80 for web traffic while keeping port 22 closed to the public. Setting up explicit NSG rules prevents unauthorized access to your application.

```
$az network nsg rule create --resource-group MyResourceGroup --nsg-name MyNSG --name AllowHTTP --pric
```

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Connecting Two VNets with Peering

VNet Peering links two separate Azure Virtual Networks together so servers in both networks can communicate privately. It works securely across the Microsoft backbone network with low latency and high speed. This is perfect when your frontend app in VNet A needs to talk securely to a database in VNet B without using public IP addresses. Peering must be configured on both sides to complete the connection.

```
$az network vnet peering create --resource-group MyResourceGroup --name VNetA-to-VNetB --vnet-name VN
```

See Key Takeaways on next page >> linkedin.com/in/mramanrajsingh

Key Takeaways

+ Always plan your IP address ranges in advance to avoid overlaps.
+ Keep database servers in private subnets without public IP addresses.
+ Group security rules into reusable Network Security Groups.
+ Use lower priority numbers (like 100) for critical rules so they get checked first.
+ Use Azure CLI scripts to build networks consistently across environments.

Watch Out For

! Overlapping IP address ranges between VNets, which breaks VNet Peering.
! Leaving port 22 (SSH) or 3389 (RDP) open to the entire internet (0.0.0.0/0).
! Forgetting that VNet Peering needs to be set up in both directions.
! Attaching public IP addresses directly to sensitive backend servers.

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